Type MS · Adding binary numbers and overflow

Answer 1 of 4 · Question 2 part c · 2 marks

2022 JUN · Paper 01 · extract 1 of 3 · find part c in this mark scheme extract

2(a)(i)	Award up to **two** marks for: • 0100 (1) 1110 (1)	Accept any number of leading zeros.	2
2(a)(ii)	**The only correct answer is B** *A is not correct because standard ASCII uses 7 bits, 64 characters would need only 6 bits* *C is not correct because standard ASCII uses 7 bits, 256 characters would need 8 bits* *D is not correct because standard ASCII uses 7 bits, 512 characters would need 9 bits*		1
2(a)(iii)	Award up to **two** marks for a linked explanation such as: All of the major languages/symbols/characters can be represented by Unicode (1) because it uses a minimum of 16 bits/more bits/32 bits/65536 characters (1) Unicode can represent all/more characters/any language (1) whereas ASCII can only represent English/Latin/128 characters/doesn't have enough characters (1) Unicode can represent all characters (1) because it uses 16 bits/2 bytes /more bits instead of 8 bits/1 byte (1)	Accept the reverse argument Allow examples of non-latin characters	2

Question Number	Answer	Additional Guidance	Mark
2(b)	1010 1011 Award up to **two** marks for: • MSB = 1 (1) • Rest of pattern correct 010 1011 (1)		2

Question Number	Answer	Additional Guidance	Mark
2(c)	Award up to **two** marks for: • 0100 (1) 1010 (1)		2

Type MS · Adding binary numbers and overflow

Answer 1 of 4 · Question 2 part c · 2 marks

2022 JUN · Paper 01 · extract 2 of 3 · find part c in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>2(d)(i)</td><td>Award up to two marks for:

• 3 bits used for all patterns (1)/
• No pattern repeated (1)

Example:

<table><tr><th>Colour</th><th>Binary pattern</th></tr><tr><td>Green</td><td>000</td></tr><tr><td>Black</td><td>001</td></tr><tr><td>White</td><td>010</td></tr><tr><td>Red</td><td>011</td></tr><tr><td>Blue</td><td>100</td></tr></table></td><td></td><td>2</td></tr>
<tr><td>2(d)(ii)</td><td>Award one mark for each of:

• 3579 x 6128 x 32 (1)
• ÷ 8 (1)
• + 732 (1)
• ÷ (1000 x 1000) (1)

Examples

$\frac{\frac{3579 \times 6128 \times 32}{8} + 732}{1000 \times 1000}$

$\frac{((3579 \times 6128 \times 32) \div 8) + 732}{1000 \times 1000}$

$\frac{(3579 \times 6128 \times 4) + 732}{1000 \times 1000}$</td><td>• Units are not required
• Equivalent expressions are accepted
• Calculations not explicit but expressed gain the mark
• Award 1 mark for correct calculated answer of 87.73 if no other marks awarded</td><td>4</td></tr>
<tr><td></td><td></td><td>Total for question 2</td><td>15</td></tr>
</table>

Type MS · Adding binary numbers and overflow

Answer 1 of 4 · Question 2 part c · 2 marks

2022 JUN · Paper 01 · extract 3 of 3 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Adding binary numbers and overflow

Answer 2 of 4 · Question 3 part e.iii · 2 marks

2019 JUN · Paper 01 · extract 1 of 4 · find part e.iii in this mark scheme extract

3(a)(i)	• LAN / Local area network (1) Allow PAN/Personal Area Network		1

Question Number	Answer	Additional Guidance	Mark
3(a)(ii)	• WAN / Wide area network (1)		1

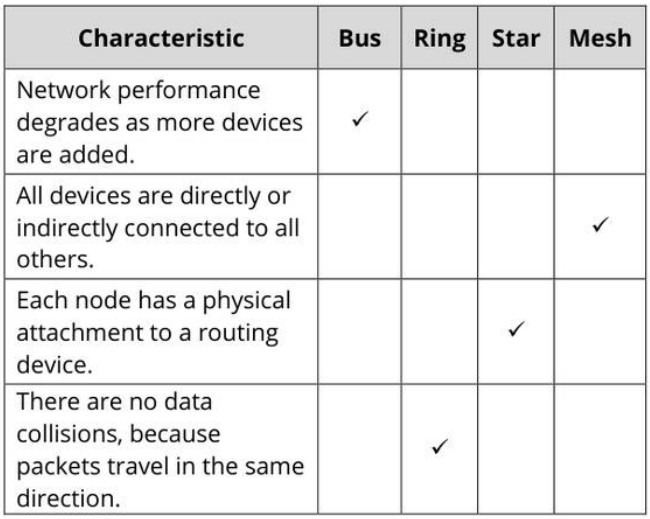
Question Number	Answer	Additional Guidance	Mark
3(b)	A: Megabits per second **The only correct answer is A** **B** is not correct because megapixels are associated with images rather than network data speeds **C** is not correct because mebicycles are not associated with network data speeds **D** is not correct because mebihertz are not associated with network data speeds		1

Type MS · Adding binary numbers and overflow

Answer 2 of 4 · Question 3 part e.iii · 2 marks

2019 JUN · Paper 01 · extract 2 of 4 · find part e.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(c)	• IMAP (1) AND **one** from: • because changes are synced (1) with the mail server in real-time (1) • because the messages always reside on the email server (1) as she has limited storage (1) on her tablet and phone • it gets sent to both smartphone and tablet (1) **Alternative answer:** • POP3 (not awarded a mark) AND **one** from: • because storage limitations on server (1) and privacy issues (1) • because more CPU time (1) may be used when syncing (1) (a large number of messages).		3

Question Number	Answer	Additional Guidance	Mark
3(d)			4

Characteristic	Bus	Ring	Star	Mesh
Network performance degrades as more devices are added.	✓			
All devices are directly or indirectly connected to all others.				✓
Each node has a physical attachment to a routing device.			✓	
There are no data collisions, because packets travel in the same direction.		✓		

Type MS · Adding binary numbers and overflow

Answer 2 of 4 · Question 3 part e.iii · 2 marks

2019 JUN · Paper 01 · extract 3 of 4 · find part e.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(e)(i)	-81 One mark for negative sign One mark for 81.		2

Question Number	Answer	Additional Guidance	Mark
3(e)(ii)	One mark for each nibble in correct order 0100 1011		2

Type MS · Adding binary numbers and overflow

Answer 2 of 4 · Question 3 part e.iii · 2 marks

2019 JUN · Paper 01 · extract 4 of 4 · find part e.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(e)(iii)	Any **two** from: 1. This is an overflow (1) error 2. It is caused by adding two 1s in the most significant bit (1) 3. The resulting number is too large to fit in the number of bits available / requires 9 bits / the number is larger than 2^8 / there is 1 that needs to be carried over (in MSB) / MSB gets discarded (1) 4. It can cause the program to crash or produce unreliable/incorrect results (1)		2

Question Number	Answer	Additional Guidance	Mark

Type MS · Adding binary numbers and overflow

Answer 3 of 4 · Question 1 part d · 2 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part d in this mark scheme extract

Number			
1(a)	**The only correct answer is B** ***A*** *is not correct because computers only understand binary* ***C*** *is not correct because Hex values will be held as binary so take up the same space* ***D*** *is not correct because Hex values are held as binary and so take the same time to execute*		1

Question Number	Answer	Additional Guidance	Mark
1(b)	Award **one** mark for each of: • 5 • A	Must be in the correct order	2

Question Number	Answer	Additional Guidance	Mark
1(c)	**The only correct answer is B** ***A*** *is not correct because this is the number of binary patterns that can be represented by 7 bits* ***C*** *is not correct because this is the number of binary patterns that can be represented by 9 bits* ***D*** *is not correct because this is the number of binary patterns that can be represented by 10 bits*		1

Question Number	Answer	Additional Guidance	Mark
1(d)	Award **one** mark for each nibble 0110 0101		2

Question Number	Answer	Additional Guidance	Mark
1(e)	Award **one** mark from: • 01000 (1) • 1000 (1)		1

Type MS · Adding binary numbers and overflow

Answer 3 of 4 · Question 1 part d · 2 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part d in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(f)	7		1

Question Number	Answer	Additional Guidance	Mark
1(g)(i)	Award **one** mark from: • The smallest block (of colour) in an image (1) • A single dot in a picture (1) • A picture element (1) • A single element of a digital screen (1)		1
1(g)(ii)	Award **one** mark from: • 8 x 9 (1) • 9 x 8 (1)	Accept 72	1
1(g)(iii)	4		1

Question Number	Answer	Additional Guidance	Mark

Type MS · Adding binary numbers and overflow

Answer 4 of 4 · Question 3 part b.iii · 2 marks

2024 JUN · Paper 01 · extract 1 of 4 · find part b.iii in this mark scheme extract

3(a)(i)	Award **one** mark for any of the following: • 256 (1) • 2^8 (1) • 2x2x2x2x2x2x2x2 (1)		1
3(a)(ii)	Award **one** mark for each of the following up to a maximum of **three** marks: • 1920 x 1080 in the numerator (1) • x 16/8 (16 in the numerator and 8 in the denominator) or (2 in the numerator) (1) • / (1024 x 1024) in the denominator (1) 1920 pixels x 1080 pixels x 8 bits / 8 bits per byte / 1024 x 1024 bytes per MiB $\frac{1920 \times 1080 \times \frac{16}{8}}{1024 \times 1024}$	Max 2 marks if not a fully correct expression. Allow equivalence e.g. $1024 = 2^{10}$ $1024^2 = 2^{20}$ 2^{-20} is equivalent to $1/2^{20}$	3
3(a)(iii)	**The only correct answer is C** ***A** is not correct because the MSB is 1 so the denary value must be more than 128* ***B** is not correct because the LSB is 0 so the denary value must be even* ***D** is not correct because the denary number is signed*		1

Type MS · Adding binary numbers and overflow

Answer 4 of 4 · Question 3 part b.iii · 2 marks

2024 JUN · Paper 01 · extract 2 of 4 · find part b.iii in this mark scheme extract

3(b)(i)	Award **one** mark for each of: • Negative sign (1) • 63 (1) Example: - 63		2
3(b)(ii)	Award **one** mark for: 1000 0010 (1)		1
3(b)(iii)	Award up to **two** marks for a linked explanation such as: • An overflow error has occurred (1) because the result requires more bits to store than are available (1)		2
3(c)(i)	Award **one** mark for any of the following: • 011 0101 (1) • 53 (1) • Hex value of 35 (1) Do not accept 11 0101 (six bits, ASCII is at least seven)		1

Type MS · Adding binary numbers and overflow

Answer 4 of 4 · Question 3 part b.iii · 2 marks

2024 JUN · Paper 01 · extract 3 of 4 · find part b.iii in this mark scheme extract

3(c)(ii)	Award **one** mark for any of the following up to a maximum of **two** marks: • It is a standard/universal encoding method (1) • It can represent all the characters in the English alphabet (1) • It only uses 7/8 bits **per character** (1) • It can represent 128/256 characters (in standard/extended format) (1) • It takes up less storage space than Unicode (1) • It can be used with bit masking (1) • It is compatible with UTF-8 (1)		2
		Total for question 3	**13**

Type MS · Adding binary numbers and overflow

Answer 4 of 4 · Question 3 part b.iii · 2 marks

2024 JUN · Paper 01 · extract 4 of 4 · find part b.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark